16S LiFePO4 BMS - External Procurement Sourcing (Non-ETStore)

Specialized Components Procured Outside ET Store (Indian Vendors: LionCircuits, Robu.in, Evelta, Rajiv Electronics)

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Sourcing Scope: Non-ETStore Vendors

LionCircuits	Robu.in	Evelta	Rajiv Electronics	Total Non-ETStore
lioncircuits.com	robu.in	evelta.com	rajivelectronics.com	4 Indian Vendors
1 Line Item | 1 pc	5 Line Items | 6 pcs	5 Line Items | 5 pcs	7 Line Items | 7 pcs (or 14 pk)	18 Line Items | 19 pcs
₹ 665.29	₹ 775.00	₹ 809.33	₹ 304.10 / ₹ 388.10	₹ 2,553.72*

#	Component & Ref	Component Description	Component Footprint	Qty	Cost Online	Total Cost	Website
Vendor: Lion Circuits (lioncircuits.com) - Primary Battery Front-End Monitoring IC							
1	**BQ76952PFBR** U1	16S LiFePO4 Battery Monitor, Autonomous Protector and Balancer IC (High-Accuracy Cell ADC)	Package_QFP:TQFP-48_7x7mm_P0.5mm	1	₹ 665.29	₹ 665.29	**lioncircuits.com**
Vendor: Robu.in (robu.in) - Current Sensing, High-Endurance Memory, Drivers and Gate Supply							
2	**FL-2B 100A 75mV** RS1	XIQI Electric CG FL-2B 100A 75mV Class 0.5 Precision DC Current Shunt (Chassis Bolt Mount)	Chassis_BoltMount (Kelvin: J31 / J32)	1	₹ 352.00	₹ 352.00	**robu.in**
3	**FM24CL64B-GTR** U21	Infineon 64 Kbit (8Kx8) High-Endurance Non-Volatile I2C FRAM (Replaces EEPROM for 10^14 Cycles)	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm	1	₹ 232.00	₹ 232.00	**robu.in**
4	**UCC27517DBVR** U2, U3	TI 4A High-Speed Low-Side Gate Drivers (Charge and Discharge Power MOSFET Switching Banks)	Package_TO_SOT_SMD:SOT-23-5	2	₹ 35.00	₹ 70.00	**robu.in**
5	**B1209S-1WR3** U14	1W 12V-to-9V Isolated DC-DC Converter Module for Gate Driver Floating Bias Rail (SIP-4 THT)	Converter_DCDC:Converter_DCDC_Murata_MEE1SxxxxSC_THT	1	₹ 109.00	₹ 109.00	**robu.in**
6	**PESD2CAN,215** D5	Nexperia Dual-Line CAN Bus Transient Surge ESD Protection TVS Diode (CANH / CANL lines)	Package_TO_SOT_SMD:SOT-23	1	₹ 12.00	₹ 12.00	**robu.in**
Vendor: Evelta (evelta.com) - Isolated CAN Transceiver, High-Voltage Buck and Precision Protection							
7	**ISO1050DUBR** U13	TI 5V Galvanically Isolated CAN Transceiver (5000 Vrms Isolation, up to 1 Mbps, SOP-8)	Package_SO:SOP-8_6.62x9.15mm_P2.54mm	1	₹ 230.10	₹ 230.10	**evelta.com**
8	**LM5164DDA** U10	TI LM5164 100V 1A Synchronous Step-Down Buck Converter (HSOP-8-EP / 7Semi Module)	Package_SO:HSOP-8-1EP_3.9x4.9mm_P1.27mm_EP2.41x3.1mm_ThermalVias	1	₹ 548.70	₹ 548.70	**evelta.com**
9	**SMBJ60CA** D3	60V 600W Surface Mount Bidirectional TVS Diode (High-Voltage Pack Input Surge Suppression)	Diode_SMD:D_SMB	1	₹ 15.00	₹ 15.00	**evelta.com**
10	**SMAJ6.0A** D20	6.0V 400W Fast Unidirectional TVS Diode (5V Isolated DC-DC Bus Rail Clamp)	Diode_SMD:D_SMA	1	₹ 4.15	₹ 4.15	**evelta.com**
11	**BCX56-135** Q21	Nexperia 80V 1A Medium-Power NPN Bipolar Transistor (Linear Pre-Regulator Pass Element)	Package_TO_SOT_SMD:SOT-89-3_Handsoldering	1	₹ 11.38	₹ 11.38	**evelta.com**

Active Electronics & Silicon Subtotal (LionCircuits, Robu.in, Evelta - 11 Items, 12 pcs): ₹ 2,249.62

16S LiFePO4 BMS - External Procurement (Rajiv Electronics & Sourcing Verification)

Interconnect Sockets, Mating Harness Cable Plugs, High-Current Terminal Hardware, and Sourcing Guidelines

#	Component & Ref	Component Description	Component Footprint	Qty	Cost Online	Total Cost	Website
Vendor: Rajiv Electronics (rajivelectronics.com) - Industrial D-Sub Connectors, Mating Harness and Hardware							
12	**DD50_Female** J30	DD-50 50-Pin D-Sub High-Density Female PCB Connector Socket (16-Cell Sense and Balance Taps)	Connector_Dsub:DSUB-50_Female_Straight	1	₹ 112.10	₹ 112.10	**rajivelectronics.com**
13	**DB25_Female** J20	DB-25 25-Pin D-Sub Female Right-Angle PCB Connector Socket (16x Cell NTC Temperature Taps)	Connector_Dsub:DSUB-25_Socket_Horizontal_P2.77x2.84mm_EdgePinOffset4.94mm	1	₹ 26.00	₹ 26.00	**rajivelectronics.com**
14	**DB50 Male Plug + Hood** P30	50-Pin D-Sub Male Solder Cable Plug with Metalized Shielded Hood (Mating Cable Harness for J30)	Cable_Mount_Dsub_50	1	₹ 95.00	₹ 95.00	**rajivelectronics.com**
15	**DB25 Male Plug + Hood** P20	25-Pin D-Sub Male Solder Cable Plug with Protective Metal Hood (Mating Cable Harness for J20)	Cable_Mount_Dsub_25	1	₹ 35.00	₹ 35.00	**rajivelectronics.com**
16	**FROM SHUNT (Bolt Terminal)** J32	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Shunt Return)	MountingHole:MountingHole_6.4mm_M6_Pad_TopBottom	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
17	**PACK MINUS (Bolt Terminal)** J33	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Pack Negative)	MountingHole:MountingHole_6.4mm_M6_Pad_TopBottom	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
18	**PACK PLUS TAP (Bolt Terminal)** J34	Direct Bolt-Down Terminal Pad (M5/M6 SS Screw, Spring/Flat Washer & Flange Nut - Pack Positive)	MountingHole:MountingHole_6.4mm_M6_Pad_TopBottom	1	₹ 12.00	₹ 12.00	**rajivelectronics.com**
GRAND TOTAL (All 18 External Line Items Across 4 Vendors)				19 pcs		**₹ 2,553.72***	**4 Vendors**

Vendor Sourcing and Fulfillment Quick Reference

Vendor Domain	Components Supplied	Stock Status	MOQ / Pack Rule	Vendor Subtotal
lioncircuits.com	TI BQ76952PFBR 16S Battery Monitor, Protector and Balancer IC (TQFP-48)	Verified In Stock	MOQ: 1 pc	₹ 665.29
robu.in	FL-2B 100A Shunt, FM24CL64 FRAM, UCC27517 Drivers (2x), B1209S DC-DC, PESD2CAN	Verified In Stock	MOQ: 1 pc	₹ 775.00
evelta.com	ISO1050 Isolated CAN Transceiver, LM5164 Buck, SMBJ60CA, SMAJ6.0A, BCX56	Verified In Stock	MOQ: 1 pc	₹ 809.33
rajivelectronics.com	DD-50 & DB-25 PCB Sockets, Mating Plugs + Hoods, M5/M6 Bolt Terminals (3x)	Verified In Stock	Design: 7 pcs | Pack: 14 pk	₹ 304.10 / ₹ 388.10*

Technical Sourcing Notes and Component Specifications

1. TI BQ76952PFBR (U1): Verified genuine TI factory stock at LionCircuits (₹665.29). Evaluated as a single bare IC in TQFP-48 package, managing autonomous cell overvoltage, undervoltage, and passive balance control.
2. XIQI Electric FL-2B Shunt Resistor (RS1): Sourced from Robu.in (₹352.00). 100A 75mV Class 0.5 precision manganese-copper current shunt. External chassis bolt-mount connected to PCB Kelvin sense header J31 and return terminal J32.
3. Infineon FM24CL64B-GTR FRAM (U21): Sourced from Robu.in (₹232.00). Replaces standard EEPROM with high-speed, non-volatile ferroelectric RAM featuring 10^14 write cycles for real-time telemetry and state-of-charge logging.
4. TI ISO1050 Galvanic CAN Transceiver (U13): Sourced from Evelta (₹230.10). Provides 5000 Vrms galvanic barrier between the BMS controller ground and the external CAN bus network, safeguarding host systems.
5. High-Current Terminals (J32, J33, J34): PCB bolt-down pads (MountingHole_6.4mm_M6_Pad_TopBottom). Stainless M5/M6 bolts, flat washers, spring washers, and flange nuts sourced from Rajiv Electronics. Hardware pack of 10 costs ₹120.00 (component design cost is ₹12.00 per set, ₹36.00 total for 3 terminals; ordering the 10-pc pack brings Rajiv subtotal to ₹388.10 and grand total to ₹2,637.72).